Web-based Element Recovery in Machine Translation

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1. Introduction

In machine translation (MT), it has been widely recognized as a challenge that source language does not express some words or phrases, which must be identified and made explicit in order to be translated into grammatical target language. For example, Chinese has no articles and marks about the countability of noun. The subject and object that can be deduced contextually by reader are also frequently omitted in Chinese sentence. However without such elements, English translation cannot be adequately generated. Although MT has made significant achievements, its quality has been known to be very poor. One of the most important reasons is their inability to recover the unexpressed elements in source language. This problem is also common to the MTs of many Asian languages to English, such as Chinese, Japanese and Korean.

In this paper, we focus on the problem of recovering unexpressed elements in Chinese in Chinese-English MT. We use the web as an additional semantic knowledge corpus for the resolutions by combining the web and the classical approaches. Due to the limited space in this paper, we just only consider two of MT problems: zero anaphora (ZA) recovery and article recovery.

2. Combination of the Web with Classical Approaches

Although the web has too many limitations to be used as an alternative to supervised models (Lapata and Keller, 2005), we suppose that the combination of the web and the classical approaches in effective ways can obtain better performance of NLP tasks. In this paper, we consider two ways of the combination.

In rule-based approaches, the web can be used as a filter to filter out low-possible candidates based on some standards, such as syntactic or semantic coherences before applying the rules to determine the most likely candidate. Such filter is supposed to be useful for improving the performance on limited rules. In statistical approaches, there have been efforts to employ the web to overcome data sparseness problem (Keller et al., 2002). The proposed approach is based on straight web models that have the limitations discussed in (Lapata and Keller, 2005). In our work, we incorporate web frequency counts as additional features into training supervised classifier to carry out the combination of the web with statistical approaches. Such web-based features are used to extract missing lexical and syntactic information in the training corpus. The two ways of the combinations are illustrated in detail by two fundamental tasks of recovering elements in Chinese-English MT: ZA and article recovery.

3. Zero Anaphora Recovery

ZAs are generally noun phrases that are understood from the context and do not need to be specified. The problem of ZA may cause considerable problems in MT because the current MTs cannot identify antecedents of ZAs. According Centering Theory, the antecedent is assumed to be the most prominent noun phrase (NP) of the receding utterance. In (Yeh and Chen, 2005), three rules have been developed based on the Centering Theory to identify the antecedents. The three rules used can deal with most of general ZAs. However Chinese is complicated and changeful language with many exceptions. The exceptions in the application of the three rules can cause misidentification. And it is very hard to produce rules exhaustively to deal with all the exceptions.

We employ the web as semantic knowledge resource based on one fact after examining the experiment examples carefully. The fact is in ZA resolution, the semantic constraint that is satisfied by a ZA, must be satisfied by its antecedent. That means that for a certain ZA, its antecedent and predicate must have semantic consistency. Therefore we employ a web-based filter to filter out the antecedent candidates with low semantic consistency before applying the identification rules. To compute the semantic consistence, two kinds of predicate-argument relations, subject-predicate and predicate-object, are considered in our work. Two patterns are constructed in the form of "NP VP" (for subject-predicate) and "VP NP" (for predicate-object). We instantiate the patterns for all antecedent candidates. These instantiated patterns are submitted to a particular web search engine (Google in our work). Our rationale is that if the consistence between ZA and a candidate is high, it is likely that the pattern instantiated by the candidate occurs frequently in the web. Therefore, filtering out candidate with low semantic consistency means to eliminate the candidate whose frequency co-occurring with the pattern in the web is low.

Our test corpus was extracted from 80 text files, about 302 utterances and 2,158 Chinese words in all, consisting of newspaper articles (current affairs, politics, economy, sports and entertainment), Chinese fictions

written by different authors and textbooks from elementary school to graduate school reading level. There are 130 ZAs in the test data and their antecedents have been annotated by hand.

We developed a simple rule-based baseline approach: for each ZA in a sentence, choose the NP in the previous sentence having the longest distance from the ZA as its antecedent. The performance is evaluated on *accuracy,* the ration of the number of correctly resolved ZAs over the number of all ZAs. The accuracies of the baseline, the approach proposed in (Yeh and Chen, 2005) and our approach are 56.9%, 74.6% and 83.1% respectively. We obtain 26.2 and 8.5 points improvements.

4. Article Recovery

Article recovery is to decide when to use *a (an)*, *the*, or *zero* article at the beginning of a noun phrase (NP). Chinese does not have any article or mark of the countability. Therefore recovering correct article in English translation is crucially important for high quality MT.

Maximum Entropy (ME) is a general technique for estimating probability distributions from data. We view article recovery as a classification problem and design a ME-based classifier to determine correct article by the probabilities of *a(an), the* and *zero* article. The primary work of modeling ME-based classifier is selection of a set of features to be considered in the analysis of the training data and estimation of the appropriate weight to each feature.

We first train and test the ME-based classifier with a set F, which consists of 10 common types of features. They are extracted from the context about NP such as the head's countability of NP, words of all words in NP, the two words left to NP, the words within the NP, POS of head. The values of the features can be obtained from the corpus using some heuristic rules. However corpus-based knowledge usually suffers from data-sparse problem. That is, some lexical and syntactic information of articles would be unseen even in a large corpus. A possible solution is the web because the size of the web is thousands of times larger than normal large corpora, and the counts obtained from the web are highly correlated with the counts from large corpora. Therefore, we developed two additional web-based features to obtain the missing lexical and syntactic information of articles from the web. The two features are described in Table 1.

For Web-F1, we estimate how frequently the NP leaded by an unknown article co-occurs with the, a/an, or zero. The value of the feature is the choice with the highest frequency in the web. For Web-F2, we take into account one word (w_1) to the left of the unknown article and the NP. The value of the feature is the article which has the highest co-occurring frequency in the web.

Table 1: Web-based features

Feature	Description	Values
Web-F1	argmax(f(art, NP))	a(an), the, zero
Web-F2	argmax(f(w_1, art, NP))	a(an), the, zero

The test corpus has a total of 921 sentences containing 2,031 NPs. All the articles of the NPs have been extracted as the correct references for the evaluation experiments. In our test corpus, the most common article is "the" article, therefore our baseline is set to 57.8% (1,174 of 2,031) at the case that "the" articles are selected for all the NPs. In order to assess the contributions of feature set F and web-based features, we built three models based on only F, the two web-based features and the combination of them respectively. The correct accuracies of the baseline and the two models based on the common features and web-based features are 57.8%, 82.7% and 83.2% respectively, while our advanced approach that combines the common and web-based features is 89.3% and significantly outperforms the other approaches.

5. Conclusion

In this paper, we employ the web as additional knowledge resource to improve the performances of the resolutions in two routine tasks of recovering unexpressed elements in Chinese-English MT. Unlike the previous researches, we show how to combine the web and the existing approaches that are rule-based and statistical approaches. Using a simple web-based filter to filter out the low-possible candidates before applying the rules based on centering theory leads to 8.5 points improvement of the accuracy in ZA resolution. In using the ME-based classifier to solving the problem of article recovery, incorporating the two web-based features into common features raises the accuracy from 82.7% to 89.3%.

References

[1] Keller, F. and Lapata, M. and Ourioupina, O. 2002. Using the Web to Overcome Data Sparseness. In Proceedings of Conference on Empirical Methods in Natural Language, 230-237.Philadelphia.

[2] Lapata, M. and Keller, F. 2005. Web-based Models for Natural Language Processing, ACM Transactions Speech and Language Processing (TSLP) 2(1), 1-31.

[3] Yeh, C. L. and Chen, Y. C. 2005. Zero Anaphora Resolution in Chinese with Shallow Parsing, Journal of Chinese Language and Computing.